# meat

Information theory

https://github.com/poyea

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# How to use this book

Each definition and each theorem identifies the Lean declaration that it describes. Each one also shows the source of that declaration. The build calculates the status of each statement from the Lean code:

- ● proved: the Lean declaration exists, and no declaration that it uses contains `sorry`.
- ◐ sorry: the statement is in Lean, but its proof or a proof that it uses is `sorry`.
- ◯ not in Lean: the statement is only in this book.
- ◇ open: the statement is an open problem. Lean states it as a `Prop`, not as a theorem.
- ✗ missing in Lean: the book identifies a Lean declaration that does not exist.

The *Uses* line shows the statements that this statement uses in Lean. Lean generates the statements of most theorems. For these theorems, the statement in this book is the statement that Lean checks.

| Chapter | ● proved | ◐ sorry | ◯ not in Lean | ◇ open | ✗ missing in Lean |
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# 1 Getting started

This chapter contains the first Lean program of the project.

**Definition 1.1** (Hello) ● proved

The constant `hello` is the string `"world"`.

```
hello                                                    Meat.lean:1
def hello := "world"
```

# Lean coverage

Each declaration in the Lean library must be in a statement of this book. All 1 declarations are in the book.